

Notes: Varela (RES, 2018)

Reallocation, Competition, and Productivity: Evidence from a Financial Liberalization Episode

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	3
2.1	Institutional setting: pre-reform capital controls in Hungary	3
2.2	What changed after liberalization	4
2.3	Main datasets	4
2.4	Main variables	4
3	Models and empirical specifications (core equations + interpretation)	5
3.1	Production, technology, and innovation costs	5
3.2	Banking sector and the financing wedge	5
3.3	Markups under Bertrand competition	5
3.4	Innovation intensities and aggregate productivity growth	6
3.5	Closed-economy equilibrium interest rate	6
3.6	Financial liberalization and the inflow case	6
3.7	Comparative statics: the five propositions	7
3.8	Baseline empirical difference-in-differences	7
3.9	R&D and innovation specification	8
3.10	Financial-dependence triple-difference design	8
3.11	Markup measurement and regressions	8
3.12	Industry-level and aggregate decompositions	9
4	Identification and empirical credibility	9
4.1	What variation identifies the effects?	9
4.2	Why the strategy is plausible	9
4.3	Why reverse causality is less worrying	9

5	Tables and figures	10
5.1	Table 1: Credit market before and after the liberalization	10
5.2	Figure 1: International borrowing of financial institutions	10
5.3	Table 2: Mean characteristics of home and foreign firms in APEH (1998)	11
5.4	Table 3: Mean characteristics of home and foreign firms in BEEPS (2001)	11
5.5	Figure 2: Evolution of main variables before and after the reform	11
5.6	Table 4: Growth rates preceding the reform	12
5.7	Table 5: Investment in capital and productivity	12
5.8	Figure 3: Effect by year	12
5.9	Table 6: R&D and innovation activities	13
5.10	Table 7: Financial dependence and investment in capital/productivity	13
5.11	Table 8: Financing terms	13
5.12	Table 9: Leverage	14
5.13	Table 10: Foreign firms' markups	14
5.14	Table 11: Markup and RTFP dispersions and industry concentration	14
5.15	Table 12: Contribution to aggregate RTFP growth	15
6	Short summary	15
7	Conclusion	16
7.1	Core conclusion (what the paper establishes)	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** How do distortions in firms' access to international capital markets affect market competition, firm-level investment in technology, and aggregate productivity growth?
- **Setting.** Hungary's 2001 deregulation of international financial flows. Before the reform, foreign-exchange controls prevented domestic firms from borrowing abroad, while foreign-owned firms could circumvent the local credit crunch using foreign funding and internal capital markets.
- **Core mechanism.** Financial liberalization reduces asymmetric access to finance across firms. This raises domestic firms' investment in technology directly (a *reallocation* channel) and also tightens product-market competition, pushing foreign firms to raise productivity as well (a *pro-competitive* channel).
- **Main prediction of the model.** In a capital-scarce economy with sufficiently developed domestic finance, lifting capital controls generates capital inflows, lowers the domestic lending rate, narrows the financing wedge between home and foreign firms, reduces foreign firms' markups, and raises aggregate productivity growth.
- **Main empirical findings.** After the reform, domestic firms experience sharply better financing terms and higher leverage; relative to foreign firms, they increase capital intensity by about 25%, labour productivity by about 5%, and RTFP by about 3%. They also become more likely to perform R&D and innovation activities. Foreign firms' markups fall by about 3%, especially in sectors with high external-finance dependence.
- **Aggregate result.** Aggregate productivity growth accelerates from 5.8% per year before the reform to 9.6% after the reform, and the source of growth shifts dramatically from reallocation toward within-firm productivity improvements.
- **Contribution.** The paper gives micro-level evidence for why financial liberalization can raise aggregate productivity: not only by reallocating funds toward constrained firms, but also by changing competitive pressure and innovation incentives for all firms.

2 Data and measurement (institutional context and key objects)

2.1 Institutional setting: pre-reform capital controls in Hungary

- **Main policy distortion.** Prior to 2001, Hungary regulated international financial flows through foreign-exchange controls. These restrictions limited banks' ability to intermediate foreign funds and, crucially, prevented most domestic firms from borrowing in foreign currency.
- **Asymmetric access to finance.** Domestic firms were largely confined to local borrowing in a shallow domestic credit market. Foreign firms, by contrast, could obtain funding abroad directly or via internal capital markets with parent companies.
- **Economic implication.** The policy created a financing wedge between domestic and foreign firms, which is the empirical analogue of the model's wedge in borrowing costs.
- **Reform.** In 2001, Act XCIII removed the remaining foreign-exchange restrictions as part of Hungary's path toward European Union accession.

2.2 What changed after liberalization

- **Capital inflows.** The liberalization was followed by large capital inflows into the banking sector and a strong expansion of local credit supply.
- **Financing terms.** By 2004, the domestic-foreign interest-rate differential fell from 3.2 percentage points to 0.65, and the collateral differential fell from 58 to 11 percentage points.
- **Credit reallocation.** Small and medium firms increased their share of total credit by 17 percentage points, and one-third of their borrowing became denominated in foreign currency.
- **Why this matters.** These patterns show that the reform genuinely relaxed the financing constraint of domestic firms rather than simply coinciding with unrelated productivity changes.

2.3 Main datasets

- **APEH.** Census balance-sheet data for all manufacturing firms in Hungary, 1992–2008, from the Statistical Department of the National Bank of Hungary.
- **Balanced panel used in main firm-level analysis.** 5,548 manufacturing firms observed continuously from 1998–2004, with at least five employees and enough information to construct productivity measures. This panel covers about 77% of manufacturing value added and 70% of manufacturing employment.
- **BEEPS.** Business Environment and Enterprise Performance Surveys (2002 and 2005), used for direct evidence on R&D, innovation, interest rates, and collateral requirements.
- **External-finance dependence.** Rajan-Zingales sector-level dependence on external finance, measured for U.S. industries and mapped into Hungarian four-digit NACE industries.

2.4 Main variables

- **Capital intensity.** Capital per worker.
- **Labour productivity.** Value added per worker.
- **Revenue productivity.** RTFP, estimated mainly using the Olley-Pakes method. The paper emphasizes that, because firm-level prices are unavailable, the object is revenue TFP, not physical TFP.
- **Markup.** Estimated as the wedge between the labour elasticity of production and the labour share.
- **Leverage.** Short-term debt over sales, available from 1999 onward for the subset of firms reporting debt to financial institutions.
- **Ownership.** A firm is classified as foreign if more than 10% of shares belong to foreign owners.
- **Innovation.** In BEEPS, a dummy for whether the firm upgraded products, adopted new technology, obtained new licensing, quality accreditation, or related innovation activities. A separate dummy records positive R&D spending.

3 Models and empirical specifications (core equations + interpretation)

3.1 Production, technology, and innovation costs

Intermediate firms produce with a Cobb-Douglas technology,

$$f(q, k, l) = qk^\alpha l^{1-\alpha}, \quad (1)$$

where q is productivity, k is capital, and l is labour.

Productivity takes a quality-ladder form,

$$q_{sj} = \lambda^{n_{sj}}, \quad \Delta_j \equiv n_{Fj} - n_{Hj}, \quad (2)$$

so Δ_j is the technology gap between foreign and home firms in variety j .

Innovation costs are

$$\Gamma(x_{Fjt}, \Delta_j) = \lambda^{-\Delta_j} \frac{1}{\phi} \frac{x_{Fjt}^2}{2}, \quad \Gamma(x_{Hjt}) = \frac{1}{\phi} \frac{x_{Hjt}^2}{2}. \quad (3)$$

- **Interpretation.** Foreign firms start ahead technologically, but home firms can leapfrog by innovating. The parameter ϕ captures the efficiency of the innovation technology, and $x_{st} \in (0, 1)$ is the innovation probability.
- **Economic idea.** Financial distortions matter because innovation requires funds before production. If the policy wedge raises domestic financing costs, domestic firms will innovate less; because competition is weaker, foreign incumbents may innovate less too.

3.2 Banking sector and the financing wedge

Banks face an intermediation cost μ , so lending and deposit rates satisfy

$$R_{t+1}^L = R_{t+1}^D + \mu. \quad (4)$$

- **Interpretation.** μ is the model's measure of domestic financial underdevelopment.
- **Capital controls.** Domestic firms cannot directly borrow abroad, while foreign firms can fund themselves at the world rate R_{t+1}^* . This creates the policy-induced financing asymmetry.

3.3 Markups under Bertrand competition

If the foreign firm is active in a variety, its markup is

$$\xi_{Fjt+1}^{post} = \frac{p_{jt+1}}{MC_{Fjt+1}^{post}} = \tau_{t+1} \lambda^{\Delta_j+1}, \quad \xi_{Fjt+1}^{pre} = \frac{p_{jt+1}}{MC_{Fjt+1}^{pre}} = \tau_{t+1} \lambda^{\Delta_j}, \quad (5)$$

where

$$\tau_{t+1} \equiv \left(\frac{R_{t+1}^L}{R_{t+1}^*} \right)^\alpha > 1. \quad (6)$$

If instead the home firm becomes the leader, its markup is

$$\xi_{Hjt+1}^{post} = \frac{p_{jt+1}}{MC_{Hjt+1}^{post}} = \frac{1}{\tau_{t+1}} \lambda. \quad (7)$$

- **Interpretation.** Foreign firms' markups rise both with technological advantage and with preferential access to cheaper capital. Home firms' markups are penalized by the financing wedge even if they successfully innovate.
- **Economic intuition.** Capital controls do not just distort quantities; they directly distort product-market power.

3.4 Innovation intensities and aggregate productivity growth

Optimal innovation intensities are

$$x_{Ft} = \frac{\phi}{\tau_{t+1}} \frac{(1 - \lambda^{-1})Y_{t+1}}{w_t R_{t+1}^*}, \quad x_{Ht} = \frac{\phi}{\tau_{t+1}^{1/\alpha}} \frac{(1 - \tau_{t+1} \lambda^{-1})Y_{t+1}}{w_t R_{t+1}^*}. \quad (8)$$

Aggregate productivity growth is

$$g_{Qt+1} = \log(\lambda)(x_{Ft} + x_{Ht}). \quad (9)$$

- **Interpretation.** Higher τ lowers innovation for both foreign and home firms. Domestic firms innovate less because post-innovation profits are squeezed by high borrowing costs. Foreign incumbents innovate less because weak competition lets them sustain high markups without needing to “escape competition.”
- **Key message.** The paper's dynamic mechanism is economy-wide: distorted access to finance depresses innovation for constrained entrants and unconstrained incumbents alike.

3.5 Closed-economy equilibrium interest rate

In the financially closed economy, the domestic deposit rate is

$$R_{t+1}^D = R_{t+1}^L - \mu = \alpha \frac{Y_{t+1}}{K_{t+1}^H} \Lambda_{t+1}^H - \mu, \quad (10)$$

where K_{t+1}^H is home firms' capital demand and Λ_{t+1}^H is an aggregate markup term for active home firms.

- **Interpretation.** Domestic rates are high when the economy is capital scarce, but low when domestic finance is underdeveloped enough to depress deposit returns.
- **Why it matters.** Whether liberalization brings capital inflows or outflows depends on the interaction between capital scarcity and domestic financial development.

3.6 Financial liberalization and the inflow case

If financial development is high enough, opening the capital account brings inflows and pins rates down at

$$\hat{R}_{t+1}^D = R_{t+1}^* + \hat{\tau}, \quad \hat{R}_{t+1}^L = R_{t+1}^* + \hat{\tau} + \mu. \quad (11)$$

- **Interpretation.** Liberalization lowers the domestic lending rate and therefore shrinks the financing wedge τ .
- **Hungary.** The empirical case lines up with this inflow regime, not the low-financial-development outflow regime.

3.7 Comparative statics: the five propositions

The model delivers five key comparative-static predictions:

$$\frac{\partial x_{Ft}}{\partial \tau_{t+1}} < 0, \quad \frac{\partial x_{Ht}}{\partial \tau_{t+1}} < 0, \quad \left| \frac{\partial x_{Ft}}{\partial \tau_{t+1}} \right| < \left| \frac{\partial x_{Ht}}{\partial \tau_{t+1}} \right|, \quad (12)$$

$$\frac{\partial (d/y)_{Hjt+1}^e}{\partial \tau_{t+1}} < 0, \quad \frac{\partial \xi_{Fjt+1}^e}{\partial \tau_{t+1}} > 0, \quad (13)$$

$$\frac{\partial \Delta_{t+1}^e}{\partial \tau_{t+1}} > 0, \quad \frac{\partial \Delta_{t+1}^e}{\partial \tau_{t+1} \partial \Delta_{t+1}} > 0, \quad \frac{\partial g_{Qt+1}}{\partial \tau_{t+1}} < 0. \quad (14)$$

- **Prediction 1.** Lower financing distortions raise innovation for both types of firms, but more so for home firms.
- **Prediction 2.** Home firms raise leverage.
- **Prediction 3.** Foreign firms' markups fall.
- **Prediction 4.** The home-foreign productivity gap narrows, especially where the initial gap was large.
- **Prediction 5.** Aggregate productivity growth rises.

3.8 Baseline empirical difference-in-differences

The basic reduced-form specification is

$$y_{it} = \delta_0 H_i + \delta_1 T_t + \delta_2 (H_i \times T_t) + \varepsilon_{it}, \quad (15)$$

where H_i is a home-firm dummy and T_t is a post-reform dummy.

The paper collapses the panel into pre- and post-reform averages, so in first differences,

$$\Delta y_i = \delta_1 + \delta_2 H_i + \Delta \varepsilon_i. \quad (16)$$

The controlled version is

$$\Delta y_{ij} = \delta_1 + \delta_2 H_i + \delta_3 Z_i + \delta_4 X_j + \delta_5 \Delta \psi_j + \Delta \varepsilon_{ij}. \quad (17)$$

- **Dependent variables.** Capital intensity, labour productivity, RTFP, markups, or leverage.
- **Controls.** Initial firm size, age, productivity, local industry pre-trends in Hungary, and global industry trends from the U.S.
- **Why first differences?** To avoid serial-correlation problems in standard DID with long firm panels.
- **Interpretation of δ_2 .** The average post-reform differential change for domestic firms relative to foreign firms.

3.9 R&D and innovation specification

Using BEEPS, the paper estimates

$$y_{ijt} = \delta_0 H_{it} + \delta_1 T_t + \delta_2 (H_{it} \times T_t) + \delta_3 Z_{it} + \mu_j + \varepsilon_{ijt}. \quad (18)$$

- **Outcomes.** Probability of positive R&D spending and probability of innovation activities.
- **Interpretation.** δ_2 measures whether domestic firms become more likely to innovate after liberalization.

3.10 Financial-dependence triple-difference design

To test whether the effects are stronger in sectors that rely more on external finance, the paper estimates

$$y_{it} = \delta_0 H_i + \delta_1 T_t + \delta_2 (H_i \times T_t) + \delta_3 (FD_j \times T_t) + \delta_4 (H_i \times FD_j \times T_t) + \delta_5 FD_j + \delta_6 (H_i \times FD_j) + \varepsilon_{it}, \quad (19)$$

and, in first differences,

$$\Delta y_{ij} = \delta_1 + \delta_2 H_i + \delta_3 FD_j + \delta_4 (H_i \times FD_j) + \delta_5 Z_i + \delta_6 X_j + \delta_7 \Delta \psi_j + \Delta \varepsilon_{ij}. \quad (20)$$

- **Interpretation of δ_3 .** How foreign firms in high-external-finance sectors respond after liberalization.
- **Interpretation of δ_4 .** Whether domestic firms expand *even more* than foreign firms in those sectors.
- **Economic logic.** High-FD sectors are exactly where pre-reform financing distortions should have hurt domestic firms most and distorted competition most.

3.11 Markup measurement and regressions

Markups are measured as

$$\xi_{ijt} = \frac{\beta_j}{\theta_{ijt}}, \quad (21)$$

where β_j is the labour elasticity of production in sector j and θ_{ijt} is the firm's labour share.

The baseline regression for foreign firms' markups is

$$\Delta \xi_{ij} = \delta_1 + \delta_2 F_i + \delta_3 Z_i + \delta_4 X_j + \delta_5 \Delta \psi_j + \Delta \varepsilon_{ij}, \quad (22)$$

and the financial-dependence version is

$$\Delta y_{ij} = \delta_1 + \delta_2 F_i + \delta_3 FD_j + \delta_4 (F_i \times FD_j) + \delta_5 Z_i + \delta_6 X_j + \delta_7 \Delta \psi_j + \Delta \varepsilon_{ij}. \quad (23)$$

- **Interpretation.** A negative δ_2 means foreign firms lose market power after the reform. A negative δ_4 means that loss is larger in the sectors where domestic firms were initially most constrained.

3.12 Industry-level and aggregate decompositions

At the industry level, the paper studies whether the dispersion gap falls more in sectors where it started larger:

$$\Delta\kappa_{jt} = \alpha + \beta_1\kappa_{jt} + \beta_2T_t + \beta_3(\kappa_{jt} \times T_t) + \varepsilon_{jt}, \quad (24)$$

where κ_{jt} is the initial foreign-home gap in RTFP, markups, or concentration.

Aggregate RTFP growth is decomposed as

$$\Delta\text{RTFP}_t = TE_t + RE_t = \sum_i D_{it}\Delta\text{RTFP}_{it} + \sum_i \sum_{z \in \{K,L\}} D_{it}(\varepsilon_{izt} - \theta_{izt})\Delta Z_{izt}. \quad (25)$$

- **Interpretation of the first term.** Within-firm technical-efficiency growth.
- **Interpretation of the second term.** Reallocation gains due to changes in the allocation of inputs across firms.
- **Why it matters.** The decomposition tells us whether post-reform productivity growth comes mainly from better allocation or from firms themselves becoming more productive.

4 Identification and empirical credibility

4.1 What variation identifies the effects?

- **Main source of variation.** Domestic and foreign firms faced very different access to international borrowing before 2001. The reform removed this asymmetry.
- **Difference-in-differences logic.** Compare domestic firms to foreign firms before and after the reform.
- **Triple-difference logic.** Interact this ownership-based exposure with sectoral dependence on external finance.

4.2 Why the strategy is plausible

- **Pre-trends.** Before 2001, domestic and foreign firms show similar trends in capital intensity, labour productivity, RTFP, and leverage; the year-by-year estimates are flat before the reform.
- **Firm controls.** Foreign firms start larger, older, more productive, and more innovative; the regressions condition on those initial differences.
- **Sector controls.** The paper controls for local industry pre-trends in Hungary and global trends from the U.S., and shows robustness to four-digit sector and sector-year fixed effects.
- **Sample-selection concern.** The author checks survival patterns and finds no significant differential survival between domestic and foreign firms around the reform.

4.3 Why reverse causality is less worrying

- **Policy timing.** The reform was part of the exogenously determined EU accession process; deregulating the capital account was the final missing requirement.

- **No obvious confounder.** Trade and FDI flows remain broadly stable around the reform, and Hungary had already undergone the major transition reforms of the 1990s.
- **Not an export shock story.** The paper shows exporters and non-exporters do not display systematically different post-reform patterns.
- **Event timing.** The yearly coefficients line up with 2001 rather than drifting earlier, which is what we would worry about if the results reflected pre-existing firm trends.

5 Tables and figures

5.1 Table 1: Credit market before and after the liberalization

Read:

- The reform is associated with a major easing in domestic credit conditions: credit-to-GDP rises from 27 to 44, the credit-to-deposit ratio rises from 83 to 113, and the lending rate falls from 12.8 to 7.5.
- For firms, the asymmetry between domestic and foreign firms shrinks sharply: the interest-rate differential drops from 3.2 to 0.65 percentage points, and the collateral differential drops from 58 to 11.
- This table is the reduced-form proof that the policy actually changed the financing environment.

TABLE 1
Credit market before and after the liberalization

Aggregate economy (%)	Before	After
Credit-to-GDP ratio	27	44
Credit-to-deposit ratio	83	113
Lending interest rate	12.8	7.5
Firms		
Credits to small and medium firms	34	51
Small and medium firms' debt in FX	0	33
Interest rate differential b. Home and Foreign	3.2	0.65
Differential in collateral b. Home and Foreign	58	11

Notes: For rows 1–5, the source is National Bank of Hungary, and data correspond to December 2000 and December 2004. Rows 6–7 come from Business Environment and Enterprise Performance Surveys of the World Bank and the European Bank for Reconstruction and Development (surveys 2002 and 2005).

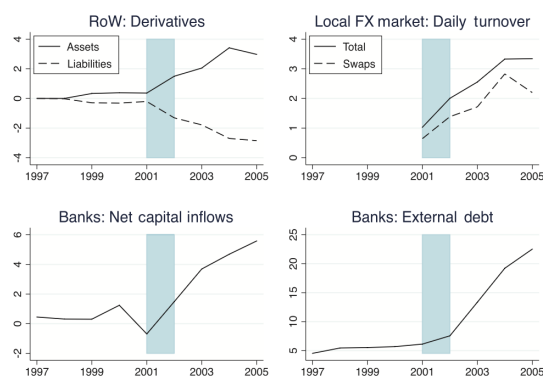


FIGURE 1
International borrowing of financial institutions

Notes: In billions of US dollars.
Source: NBH and IMF.

5.2 Figure 1: International borrowing of financial institutions

Read:

- The figure shows a clear break around 2001: derivatives activity, FX swaps, banks' net capital inflows, and banks' external debt all rise strongly after the reform.
- This visual evidence is important because the theory is explicitly about the *capital-inflow* case of financial liberalization.

- In other words, Hungary is empirically in the model regime where liberalization lowers domestic rates and reduces financing distortions.

5.3 Table 2: Mean characteristics of home and foreign firms in APEH (1998)

Read:

- Foreign firms are larger, older, more productive, and have higher markups before the reform.
- This matters because raw post-reform comparisons would be misleading without conditioning on pre-existing differences.
- The paper’s empirical design is therefore not “small versus big firms,” but “domestic versus foreign firms, conditional on initial observables and common trends.”

TABLE 2
Mean characteristics of home and foreign firms (1998): APEH database

	Foreign	Home (ln logs)	Difference in means
Value added	10.6549 (0.0525)	9.0769 (0.0226)	1.5779*** (0.0500)
Employment	3.8952 (0.0429)	2.8602 (0.0191)	1.0349*** (0.0418)
Labour productivity	6.7596 (0.0263)	6.2167 (0.0131)	0.5429*** (0.0278)
RTFP	1.4093 (0.0267)	1.1959 (0.0139)	0.2133*** (0.0291)
Markup	0.2391 (0.0159)	0.1774 (0.0098)	0.0617*** (0.0197)
Age	1.6167 (0.0136)	1.4777 (0.0090)	0.1390*** (0.0179)
Number of firms	1,283	4,165	5,448

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors in parentheses.
Source: APEH.

TABLE 3
Mean characteristics of home and foreign firms (2001): BEEPS database

	Foreign	Home	Difference in means
Probability of innovation	0.5946 (0.0818)	0.3521 (0.0328)	0.2425*** (0.0858)
Probability of R&D	0.3206 (0.0647)	0.1675 (0.0267)	0.1532*** (0.0614)
Interest rate paid	9.0667 (0.9200)	13.3198 (0.5845)	-4.2531*** (1.2687)
Required value of collateral	124.2105 (13.7504)	185.2874 (11.5619)	-61.0768*** (25.6236)
Number of firms	53	197	250

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors in parentheses.
Source: BEEPS.

5.4 Table 3: Mean characteristics of home and foreign firms in BEEPS (2001)

Read:

- Before the reform, foreign firms are more likely to conduct innovation and R&D, and they face much better borrowing terms.
- Domestic firms pay about 4.25 percentage points more in interest and post substantially more collateral.
- This is direct micro evidence that the pre-2001 distortion is genuinely a financing distortion, not just a productivity gap.

5.5 Figure 2: Evolution of main variables before and after the reform

Read:

- Before 2001, home and foreign firms move broadly in parallel. After 2001, domestic firms pull away in labour productivity, RTFP, capital intensity, and leverage.
- Foreign firms’ markups fall more sharply after the reform.
- This figure visually previews the whole paper: finance improves for domestic firms, domestic productivity rises, and competition intensifies.

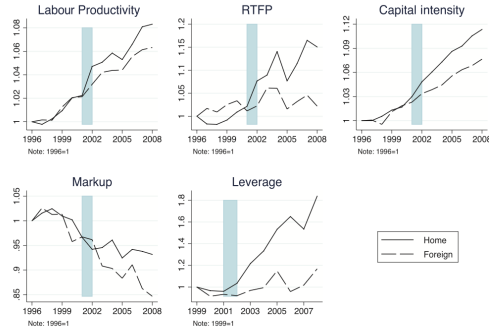


FIGURE 2
Evolution of main variables before and after the reform

TABLE 4
Growth rates preceding the reform

Balanced panel	Home	Foreign	Difference in means
Capital intensity	0.0235 (0.0032)	0.0289 (0.0040)	-0.0054 (0.0061)
Labour productivity	0.0554 (0.0043)	0.0697 (0.0074)	-0.0143 (0.0087)
RTFP	0.0264 (0.0041)	0.0395 (0.0071)	-0.0132 (0.0082)
Markup	-0.0076 (0.0040)	0.0058 (0.0068)	-0.0133* (0.0080)
Leverage	-0.0077 (0.0345)	0.0364 (0.0644)	-0.0441 (0.0692)
N	17,765	5,654	23,419

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors in parentheses. The table reports the mean of the variable growth rate within the five years prior to the reform (1996–2000).
Source: APEH.

5.6 Table 4: Growth rates preceding the reform

Read:

- The pre-reform differences in growth rates are small and mostly statistically insignificant for capital intensity, labour productivity, RTFP, and leverage.
- This is the formal support for the parallel-trends assumption suggested visually in Figure 2.
- It strengthens the DID interpretation of the post-2001 divergence.

5.7 Table 5: Investment in capital and productivity

Read:

- Domestic firms expand after the reform: with full controls, capital intensity rises by about 0.252 log points, labour productivity by about 0.053, and RTFP by about 0.032.
- These are the paper's baseline firm-level reallocation results.
- The stability of coefficients across specifications shows the finding is not driven by initial firm differences or sector trends.

TABLE 5
Investment in capital and productivity

	Δ Capital intensity			Δ Labour productivity			Δ RTFP		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Home	0.239*** (0.023)	0.253*** (0.025)	0.252*** (0.025)	0.074*** (0.017)	0.051*** (0.017)	0.053*** (0.016)	0.098*** (0.015)	0.032** (0.014)	0.032** (0.016)
Firm-level controls		Yes	Yes		Yes	Yes		Yes	Yes
Local trends			Yes		Yes	Yes		Yes	Yes
Global trends			Yes		Yes	Yes		Yes	Yes
R ²	0.019	0.030	0.030	0.004	0.027	0.040	0.008	0.075	0.088
N	5,448	5,448	5,448	5,448	5,448	5,448	5,448	5,448	5,448

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors are clustered at four-digit NACE industries. All regressions include a constant term. Global industry controls include capital intensity and TFP growth rates of the four-digit NACE industries in the U.S. between 1998 and 2004. Local industry controls are capital intensity and RTFP average growth rates at the four-digit level in Hungary in the late 90s. Firm-level controls are age, employment and RTFP in the initial year (1998).
Source: APEH.

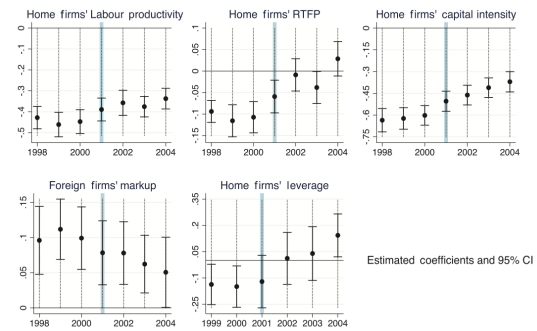


FIGURE 3
Effect by year

5.8 Figure 3: Effect by year

Read:

- The yearly interaction coefficients are roughly flat before 2001 and then move monotonically after the reform.
- This timing result is essential: it ties the domestic-firm expansion and foreign-markup decline directly to liberalization.
- It also acts as a placebo test against stories based on pre-existing differential trends.

5.9 Table 6: R&D and innovation activities

Read:

- Domestic firms become more likely to undertake R&D and innovation after the reform.
- In the fully controlled regressions, the reform raises the probability of R&D by about 9 percentage points and innovation by about 12 percentage points.
- This matters because the paper's mechanism is not only about scale or factor deepening; it is specifically about technology investment.

	R&D activities			Innovation activities		
	(1)	(2)	(3)	(4)	(5)	(6)
Home	-0.153*** (0.028)	-0.058 (0.032)	-0.032 (0.030)	-0.242*** (0.057)	-0.158** (0.054)	-0.090 (0.056)
Home*Reform	0.107* (0.048)	0.083** (0.033)	0.090* (0.044)	0.176** (0.066)	0.167** (0.055)	0.122* (0.056)
Reform	0.023 (0.055)	0.046 (0.052)	0.023 (0.043)	-0.084 (0.063)	-0.071 (0.075)	-0.099 (0.081)
Firm-level controls		Yes	Yes		Yes	Yes
Sector-fixed effects			Yes			Yes
R ²	0.019	0.064	0.081	0.014	0.037	0.069
N	774	774	774	774	774	774

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors are clustered at industry level. All regressions include a constant term. R&D is a dummy if the firm reports positive R&D expenditures. Innovation is a dummy if a firm reports any of the following activities: successfully developed a major product line, upgraded an existing product line, acquired a new production technology, obtained a new licensing agreement, and obtained a new quality accreditation. Firm-level controls are age and size.
Source: BEEPS.

	Δ Capital intensity			Δ Labour productivity			Δ RTFP		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Home	0.210*** (0.021)	0.221*** (0.024)	0.219*** (0.024)	0.058*** (0.017)	0.017 (0.015)	0.015 (0.017)	0.083*** (0.018)	-0.010 (0.023)	-0.016 (0.023)
Home * Fin. dep.	0.142* (0.080)	0.156* (0.076)	0.155* (0.077)	0.093* (0.053)	0.155*** (0.045)	0.147*** (0.046)	0.087 (0.072)	0.181** (0.080)	0.167** (0.067)
Fin. dep.	-0.084 (0.064)	-0.061 (0.070)	-0.053 (0.077)	0.276** (0.124)	0.320** (0.124)	0.334** (0.134)	0.162 (0.107)	0.222** (0.10)	0.277*** (0.092)
Firm-level controls		Yes	Yes		Yes	Yes		Yes	Yes
Local trends			Yes		Yes	Yes		Yes	Yes
Global trends			Yes		Yes	Yes		Yes	Yes
R ²	0.020	0.031	0.031	0.034	0.074	0.081	0.022	0.111	0.120
N	5,143	5,143	5,143	5,143	5,143	5,143	5,143	5,143	5,143

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors are clustered at four-digit NACE industries. All regressions include a constant term. Financial Dependence is the Rajan and Zingales' (1998) index. Global industry controls include capital intensity and TFP growth rates of the four-digit NACE industries in the U.S. between 1998 and 2004. Local industry controls are capital intensity and RTFP average growth rates at the four-digit level in Hungary in the late 90s. Firm-level controls are age, employment and RTFP in the initial year (1998).
Source: APEH.

5.10 Table 7: Financial dependence and investment in capital/productivity

Read:

- Domestic firms expand more in sectors that depend more on external finance. This is exactly what we should expect if the reform works through financing constraints.
- Foreign firms in those sectors also raise labour productivity and RTFP, but not capital intensity. That pattern is crucial: it is consistent with tighter competition, not with a pre-reform credit constraint on foreign firms.
- This is the paper's cleanest evidence separating the *reallocation* channel from the *pro-competitive* channel.

5.11 Table 8: Financing terms

Read:

- Domestic firms borrow at worse terms before the reform, but the home-foreign gap shrinks dramatically afterward.
- In the fully controlled regressions, the domestic interest-rate penalty falls by about 3.95 percentage points and the collateral penalty by about 31 percentage points.
- This table links the productivity results back to the actual financing mechanism rather than leaving it inferred.

TABLE 8
Financing terms

	Interest rate			Value of collateral		
	(1)	(2)	(3)	(4)	(5)	(6)
Home	4.253*** (1.132)	3.707*** (1.027)	3.729*** (1.051)	60.789*** (15.391)	49.174** (15.727)	52.106*** (11.263)
Home*Reform	-3.879** (1.134)	-3.858*** (1.018)	-3.947*** (1.076)	-37.653* (17.130)	-35.438* (17.104)	-31.170** (10.911)
Reform	-0.026 (0.951)	-0.159 (0.830)	-0.221 (0.890)	20.968 (12.571)	19.574 (13.192)	13.368 (11.635)
Firm-level controls		Yes	Yes		Yes	Yes
Sector- fixed effects			Yes			Yes
R ²	0.175	0.202	0.217	0.035	0.045	0.103
N	415	415	415	399	399	399

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors are clustered at industry level. All regressions include a constant term. Firm-level controls are age and size.
Source: BEEPS.

TABLE 9
Leverage

	Δ Leverage			
	(1)	(2)	(3)	(4)
Home	0.160** (0.073)	0.239*** (0.085)	0.230*** (0.088)	0.238** (0.100)
Home* Financial dependence				0.526** (0.266)
Financial dependence				-0.595** (0.234)
Firm-level controls		Yes	Yes	Yes
Local trends			Yes	Yes
Global trends			Yes	Yes
R ²	0.002	0.006	0.007	0.015
N	2,742	2,742	2,742	2,742

Notes: *, **, *** significant at the 10, 5, and 1 percent level. Standard errors are clustered at four-digit NACE industries. All regressions include a constant term. Financial dependence is the Rajan and Zingales (1998) index. Global industry controls include capital intensity and TFP growth rates of the four-digit NACE industries in the U.S. between 1998 and 2004. Local industry controls are capital intensity and RTFP average growth rates at the four-digit level in Hungary in the late 90s. Firm-level controls are age, employment and RTFP in the initial year (1998).
Source: APEH.

5.12 Table 9: Leverage

Read:

- Domestic firms' leverage rises after the reform, and the increase is strongest in sectors with high external-finance dependence.
- Foreign firms do not raise leverage in those sectors; if anything, the coefficient goes in the opposite direction.
- This fits the idea that credit is reallocated toward previously constrained domestic firms.

5.13 Table 10: Foreign firms' markups

Read:

- Foreign firms' markups decline after the reform: about 0.026 log points in the controlled baseline.
- The decline is larger in sectors with higher financial dependence, where pre-reform competitive distortions were worst.
- This is the central reduced-form evidence for the pro-competitive effect.

5.14 Table 11: Markup and RTFP dispersions and industry concentration

Read:

- At the industry level, the foreign-home gaps in RTFP and markups shrink more in sectors where they were initially larger.

TABLE 10
Foreign firms' markups

	Δ Markups			
	(1)	(2)	(3)	(4)
Foreign	-0.017*	-0.025**	-0.026**	0.030*
	(0.009)	(0.011)	(0.012)	(0.016)
Foreign*Financial dependence				-0.205***
				(0.043)
Financial dependence				0.212***
				(0.069)
Firm-level control		Yes	Yes	Yes
Local trend			Yes	Yes
Global trends			Yes	Yes
R^2	0.000	0.023	0.024	0.057
N	5,376	5,376	5,376	5,086

Notes: *, **, *** significant at the 10%, 5%, and 1% level. Standard errors are clustered at four-digit NACE industries. All regressions include a constant term. Financial dependence is the Rajan and Zingales (1998) index. Global industry controls include capital intensity and TFP growth rates of the four-digit NACE industries in the U.S. between 1998 and 2004. Local industry controls are capital intensity and RTFP average growth rates at the four-digit level in Hungary in the late 90s. Firm-level controls are age, employment and RTFP in the initial year (1998).
Source: APEH.

TABLE 11
Markup and RTFP dispersions and industry concentration

	Change in RTFP dispersion			Change in markup dispersion			Change in concentration		
	Late 90s	Reform	Accounting for Pre-trends	Late 90s	Reform	Accounting for Pre-trends	Late 90s	Reform	Accounting for Pre-trends
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Initial value	-0.076	-0.202**	-0.076	-0.419***	-0.730***	-0.419***	-0.177***	-0.317***	-0.177***
	(0.064)	(0.079)	(0.077)	(0.063)	(0.135)	(0.079)	(0.075)	(0.085)	(0.060)
Initial value*T			-0.222**			-0.310**			-0.245***
			(0.107)			(0.140)			(0.091)
T			0.186			0.134**			0.211***
			(0.128)			(0.054)			(0.072)
R^2	0.018	0.074	0.100	0.354	0.280	0.325	0.101	0.145	0.223
N	82	82	164	78	78	156	82	82	164

Notes: All regressions include a constant term. *, **, *** significant at the 10%, 5%, and 1% level. Standard errors in parentheses. Three-digit NACE industries correlations.
Source: APEH.

TABLE 12
Contribution to aggregate RTFP growth

	Total sample		Balanced panel	
	Δ RTFP	Reallocation	Within-firm	Within-firm
	(1)	(2)	(3)	(4)
A: Mean Growth Rate				
Before	5.8	4.8	1.0	0.9
After	9.6	1.7	7.9	7.3
B: Contribution to aggregate RTFP growth (wrt column 1)				
Before	100.0	83.5	16.5	16.5
After	100.0	18.0	82.0	75.4

Notes: Values in %.
Source: APEH.

- Industry concentration also falls more in sectors that started more concentrated.
- This is the sector-level version of the same story: domestic firms catch up and market power becomes less concentrated.

5.15 Table 12: Contribution to aggregate RTFP growth

Read:

- Aggregate productivity growth rises from 5.8% before the reform to 9.6% after it.
- Before the reform, growth is mostly reallocation-driven: 4.8 percentage points out of 5.8. After the reform, growth is mostly within-firm: 7.9 percentage points out of 9.6.
- This is the paper's aggregate punchline. Liberalization does not merely reassign resources; it changes firms' own incentives to raise productivity.

6 Short summary

- **Conclusion.** Hungary's financial liberalization reduced policy-created financing asymmetries between domestic and foreign firms and raised aggregate productivity.
- **Intuition.** Better credit conditions let constrained domestic firms invest more, while stronger competition forced foreign incumbents to raise productivity and accept lower markups.
- **Empirical lesson.** The productivity effects of financial liberalization can come as much from competition and innovation incentives as from static capital reallocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Capital controls can distort competition by giving some firms privileged access to finance. Removing those controls can raise productivity through both easier financing for constrained firms and tougher competition for unconstrained firms.
- **Magnitude.** The effects are economically large: domestic firms increase leverage, capital intensity, labour productivity, RTFP, and innovation activity, while foreign firms' markups fall and aggregate productivity growth accelerates sharply.
- **Credibility.** The empirical design exploits a policy-created pre-reform asymmetry, documents flat pre-trends, ties the effects to 2001 timing, and shows stronger responses exactly where the financial channel predicts them: sectors with high external-finance dependence.
- **Aggregate implication.** The post-reform growth acceleration is driven mainly by within-firm productivity gains, not just by reallocation across existing firms.

7.2 Economic intuition

- Financial distortions do more than starve constrained firms of capital. They also soften competition by protecting incumbents from aggressive challengers.
- When domestic firms suddenly gain access to cheaper funds, they invest, innovate, and expand. That threatens foreign incumbents, compresses their markups, and induces them to innovate more as well.
- The key amplification mechanism is therefore product-market competition: relaxing a financing distortion changes the innovation incentives of the whole industry, not only of the previously constrained firms.

7.3 Takeaway

This paper is a strong example of how finance, competition, and productivity are jointly determined. Its main insight is that financial liberalization can raise productivity not only because “more credit” reaches the right firms, but because removing asymmetric access to credit changes who competes, how hard they compete, and how much everyone invests in technology. For development and international macro, that is the central lesson: capital controls can create micro distortions whose aggregate cost shows up through lower competition and slower within-firm productivity growth.